

Linyi (Tiger) Hou

✉ linyh2@illinois.edu | github.com/TigerHou2 | U.S. Permanent Resident

EDUCATION

Ph.D. in Aerospace Engineering

May 2025

University of Illinois Urbana-Champaign

GPA: 3.94/4.00

Thesis: Autonomous Lost in Space and Time State Determination using Periodic Variable Stars

B.S. in Aerospace Engineering, Minor in Computer Engineering

May 2021

University of Illinois Urbana-Champaign

GPA: 3.93/4.00

SKILLS

Programming: Python (numpy | scipy | astropy | spiceypy | matplotlib | pybind11), C++ (Eigen | OpenMP), MATLAB (Simulink), Javascript (NodeJS), Julia, ADQL, Slurm

Software Development: Git, GitHub, continuous integration, unit testing

Analysis: Monte Carlo simulation, state estimation, navigation system modeling, orbit determination

Collaboration: Trello, technical/proposal writing, MS Office suite, L^AT_EX

EXPERIENCE

Graduate Researcher, University of Illinois Urbana-Champaign

Jun 2021 – May 2025

- Developed Python tools to model the astrometry and photometry of 300+ variable stars and 1000+ asteroids; validated timing accuracy against the pulsar timing package PINT to within 50 μ s.
- Developed a novel initial orbit determination technique in MATLAB using batched sequential range-rate measurements that achieved over 10x better accuracy than state-of-the-art methods.
- Developed a star identification technique using C++ which is robust against stellar aberration effects at relativistic speeds, improving star identification success rate from 38% to 91%.

Teaching Assistant, University of Illinois Urbana-Champaign

Jun 2021 – May 2024

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and develop code and instructional materials for courses in flight mechanics, systems engineering, and quadcopter controls.
- Helped students debug quadcopter flight control firmware (C) and software (Python), diagnose and fix hardware issues in a class with over 100 students.

Software Engineering Intern, Collins Aerospace

May 2019 – Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ for modern OS compatibility.
- Created a GUI application for the aforementioned avionics data interpreter using Python.

PROJECTS

Personal Projects

- Developing an open source Python package that generates star tracker images based on physical modeling of camera components, stellar spectra, and relativistic effects.
- Used Javascript, HTML, and CSS to create interactive puzzles for a puzzle hunt website.
- Created a two-player videogame with destructible terrain on an FPGA using System Verilog.

- Wrote C/Arduino scripts to handle input mapping and battery charging for a 3D-printed, life-size, remote-controlled R2D2.

Collaborative Projects in Flight Dynamics

2022, 2023

- Planetary entry: Created a 3-DOF entry simulation in MATLAB to assess the flight performance of several novel Apollo-derived entry guidance algorithms at Mars.
- Trajectory optimization: Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories in MATLAB.

Team Lead/Advisor, NASA's RASC-AL Competition

2019, 2022, 2023

- Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.).
- Advised two student teams who developed technical requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.

HONORS AND AWARDS

<i>Aerospace Engineering Alumni Advisory Board Fellowship</i> , UIUC	2024
<i>Best Student Paper</i> , 46 th AAS GN&C Conference, American Astronautical Society	2024
<i>Yee Memorial Fellowship</i> , UIUC	2021, 2024
<i>David and Catherine Thompson Space Technology Scholarship</i> , AIAA	2020

SELECT PUBLICATIONS

JOURNAL ARTICLES

- [J.3] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars". In: *The Journal of the Astronautical Sciences* 72.9 (2025). DOI: [10.1007/s40295-025-00485-8](https://doi.org/10.1007/s40295-025-00485-8).
- [J.2] **L. Hou** and Zachary R Putnam. "Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements". In: *The Journal of the Astronautical Sciences* 71.24 (2024). DOI: [10.1007/s40295-024-00442-x](https://doi.org/10.1007/s40295-024-00442-x).
- [J.1] **L. Hou** and Zachary R Putnam. "A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV". In: *The Journal of the Astronautical Sciences* 71.6 (2024). DOI: [10.1007/s40295-023-00425-4](https://doi.org/10.1007/s40295-023-00425-4).

CONFERENCE PRESENTATIONS

- [C.3] **L. Hou** and Siegfried Eggl. "Autonomous Optical Navigation using Astrometry and Photometry". In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua'i, HI, Jan. 2025.
- [C.2] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars". In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.
- [C.1] **L. Hou**, Kevin Lohan, and Zachary R Putnam. "Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms". In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.